

QUANTUM MECHANICS | ADVANCED UNDERGRADUATE PHYSICS

Photon Polarization as a Spin-1 Quantum System

Helicity States, Basis Transformations, and Matrix Mechanics

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Level: Junior/Senior Undergraduate
(PHY 341) | 50–60 MinutesPrereqs: Dirac Notation, Rotation
Operators, Unitary Matrices, Wave
Optics

Artifact: Instructional Sample

Central Idea: Connecting classical polarization optics to quantum Hilbert space, highlighting how operator matrices depend on basis while physical observables remain invariant.

Instructional Focus & Learning Objectives

The study of photon polarization provides one of the cleanest pedagogical bridges between classical electromagnetism and formal quantum mechanics. In classical optics, polarizing filters and wave plates manipulate transverse electromagnetic waves. In quantum mechanics, these same optical elements represent projection operators and unitary basis transformations acting on single-photon quantum states.

This lesson explores how the photon—an intrinsic spin-1 gauge boson—is modeled in quantum mechanics, carefully distinguishing between massive spin-1 systems and massless relativistic particles, developing the unitary transformation between linear and circular polarization bases, and calculating operator matrix representations.

- **Physical Rigor:** Understand why a massless spin-1 photon propagating in free space possesses only *two* physical helicity states ($\lambda = \pm 1$), in sharp contrast to massive spin-1 particles which possess three projection states ($m_s = +1, 0, -1$).
- **Basis Transformation:** Construct the unitary transformation matrix U connecting the linear polarization basis $\{|x\rangle, |y\rangle\}$ to the circular polarization eigenbasis $\{|R\rangle, |L\rangle\}$.
- **Matrix Representation of Operators:** Apply similarity transformations to calculate the matrix representation of the angular momentum operator $\hat{J}_z$ in the linear basis, verifying that physical eigenvalues are preserved across basis changes.
- **Optics–Quantum Connection:** Model optical elements (polarizers, quarter-wave plates) as quantum operators, and examine how two-level polarization states form the foundation of photon qubits and Bell-state entanglement.

Physical Foundations: Massive Spin-1 vs. Massless Photons

The Helicity of Massless Gauge Bosons

For a **massive** spin-1 particle (such as a ρ meson, deuteron, or W/Z boson), there exists a rest frame. The spin angular momentum projection along any axis z has $2s + 1 = 3$ allowed quantum numbers:

$$m_s \in \{+1, 0, -1\} \implies S_z \in \{+\hbar, 0, -\hbar\}$$

For a **massless** particle like the photon ($m = 0$), no rest frame exists; the photon travels at the speed of light c in all inertial frames. Relativistic wave equations and electromagnetic gauge invariance ($\nabla \cdot \mathbf{E} = 0$ in free space) enforce strict transversality: the electric and magnetic fields must be purely perpendicular to the propagation vector $\mathbf{k}$.

Consequently, for a photon propagating along the z -axis, the longitudinal state corresponding to $m_s = 0$ is strictly forbidden by gauge symmetry. A propagating photon exhibits only **two physical polarization states**:

- **Right-Circular Polarization** $|R\rangle$: Helicity $\lambda = +1$, carrying angular momentum $J_z = +\hbar$.
- **Left-Circular Polarization** $|L\rangle$: Helicity $\lambda = -1$, carrying angular momentum $J_z = -\hbar$.

Thus, the physical polarization Hilbert space of a single propagating photon is a **two-dimensional complex vector space** ($\mathbb{C}^2$), isomorphic to a spin-1/2 system and forming a natural physical qubit.

Anticipated Misconception: The “Missing” Third State

Misconception: Students often ask: “If a photon has spin $s = 1$, why isn’t its polarization described by a 3×3 matrix like the spin-1 Stern–Gerlach experiment?”

Clarification: Remind students that Wigner’s classification of the Poincaré group dictates that massless particles are classified by *helicity* (the projection of angular momentum along momentum $\mathbf{p} / |\mathbf{p}|$), not by rest-frame spin. The absence of mass eliminates the longitudinal $m = 0$ degree of freedom in free radiation. In quantum optics, photon polarization is strictly a 2D two-level system.

Two Orthonormal Bases for Polarization

A beam of monochromatic light propagating in the $+z$ direction can be decomposed into two distinct, complete orthonormal bases spanning the 2D transverse polarization space.

1. The Linear Polarization Basis: $\{|x\rangle, |y\rangle\}$

Representing photons linearly polarized along the horizontal (x) and vertical (y) axes:

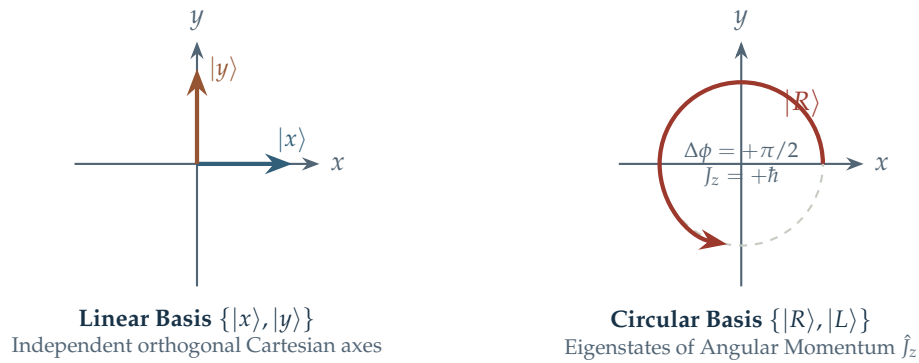
$$|x\rangle \longleftrightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |y\rangle \longleftrightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (1)$$

Orthonormality holds: $\langle x|x\rangle = \langle y|y\rangle = 1$ and $\langle x|y\rangle = 0$.

2. The Circular Polarization Basis: $\{|R\rangle, |L\rangle\}$

Right- and left-circularly polarized light correspond to electric field vectors rotating in the transverse xy -plane. In the quantum state space, circular states are constructed from linear superpositions with a relative phase shift of $\pm\pi/2$ ($\pm i$):

$$|R\rangle = \frac{1}{\sqrt{2}}(|x\rangle + i|y\rangle), \quad |L\rangle = \frac{1}{\sqrt{2}}(|x\rangle - i|y\rangle) \quad (2)$$



Phase Convention Commentary

Note on Sign Conventions:

In optics textbooks, definitions of right- and left-circular polarization depend on whether the observer looks toward the source or along the propagation direction. In quantum mechanics (following Townsend, Section 2.5), we define $|R\rangle$ such that under a spatial rotation about the z-axis by angle ϕ , $|R\rangle$ acquires a phase $e^{-i\phi}$ and $|L\rangle$ acquires $e^{+i\phi}$. This establishes that $|R\rangle$ is the eigenstate of $\hat{J}_z$ with eigenvalue $+\hbar$, and $|L\rangle$ is the eigenstate with $-\hbar$.

Optical Realization: The Quarter-Wave Plate

How do we experimentally transform linear states into circular eigenstates? In the laboratory, this is accomplished via a birefringent crystal called a **Quarter-Wave Plate (QWP)**.

Action of a Quarter-Wave Plate

A quarter-wave plate has two orthogonal optical axes: a “fast axis” and a “slow axis”. The crystal thickness is engineered so that light polarized along the slow axis experiences a phase retardation of exactly $\pi/2$ (a quarter cycle, $e^{i\pi/2} = i$) relative to the fast axis.

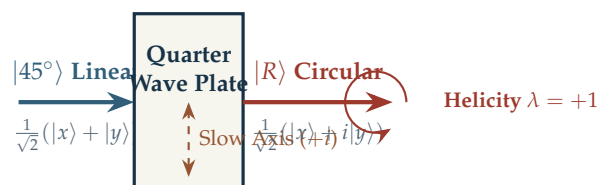
If linear light polarized at an angle of 45° to the axes enters the plate:

$$|45^\circ\rangle = \frac{1}{\sqrt{2}}(|x\rangle + |y\rangle)$$

Orienting the slow axis along the y -axis imparts a relative phase factor of $+i$ to the $|y\rangle$ component:

$$|45^\circ\rangle \xrightarrow{\text{QWP}} \frac{1}{\sqrt{2}}(|x\rangle + i|y\rangle) = |R\rangle$$

Thus, a quarter-wave plate physically converts a linear superposition into a pure angular-momentum eigenstate $|R\rangle$.



Worked Example: Matrix Representation of $\hat{J}_z$ in Both Bases

Problem Statement (Townsend, Example 2.8, p. 65)

Determine the matrix representation of the angular momentum operator $\hat{J}_z$ using both:

- (a) The circular polarization basis $\{|R\rangle, |L\rangle\}$
- (b) The linear polarization basis $\{|x\rangle, |y\rangle\}$

Part (a): Matrix Representation in the Circular Basis $\{|R\rangle, |L\rangle\}$

By construction, $|R\rangle$ and $|L\rangle$ are the physical eigenstates of $\hat{J}_z$:

$$\hat{J}_z|R\rangle = +\hbar|R\rangle, \quad \hat{J}_z|L\rangle = -\hbar|L\rangle$$

The matrix elements are given by the projection $\langle i|\hat{J}_z|j\rangle$:

$$\begin{aligned} \langle R|\hat{J}_z|R\rangle &= +\hbar, & \langle R|\hat{J}_z|L\rangle &= 0 \\ \langle L|\hat{J}_z|R\rangle &= 0, & \langle L|\hat{J}_z|L\rangle &= -\hbar \end{aligned}$$

Because the circular basis is the eigenbasis of $\hat{J}_z$, the matrix is strictly diagonal:

$$[\hat{J}_z]_{\text{circ}} = \begin{pmatrix} \hbar & 0 \\ 0 & -\hbar \end{pmatrix} = \hbar \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \hbar \hat{\sigma}_z \quad (3)$$

Part (b): Transformation to the Linear Basis $\{|x\rangle, |y\rangle\}$

To find $[\hat{J}_z]_{\text{lin}}$, we construct the unitary change-of-basis matrix U that transforms states from the circular basis to the linear basis:

$$U = \begin{pmatrix} \langle x|R\rangle & \langle x|L\rangle \\ \langle y|R\rangle & \langle y|L\rangle \end{pmatrix}$$

From the definitions $|R\rangle = \frac{1}{\sqrt{2}}(|x\rangle + i|y\rangle)$ and $|L\rangle = \frac{1}{\sqrt{2}}(|x\rangle - i|y\rangle)$, the inner products are:

$$\langle x|R\rangle = \frac{1}{\sqrt{2}}, \quad \langle x|L\rangle = \frac{1}{\sqrt{2}}, \quad \langle y|R\rangle = \frac{i}{\sqrt{2}}, \quad \langle y|L\rangle = -\frac{i}{\sqrt{2}}$$

Hence, the transformation matrix U and its conjugate transpose $U^\dagger$ are:

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}, \quad U^\dagger = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} \quad (4)$$

Check unitarity: $UU^\dagger = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I.$

Now apply the similarity transformation:

$$[\hat{J}_z]_{\text{lin}} = U[\hat{J}_z]_{\text{circ}}U^\dagger$$

First, compute the product $U[\hat{J}_z]_{\text{circ}}$:

$$U[\hat{J}_z]_{\text{circ}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} \begin{pmatrix} \hbar & 0 \\ 0 & -\hbar \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \hbar & -\hbar \\ i\hbar & i\hbar \end{pmatrix}$$

Next, multiply on the right by $U^\dagger$:

$$\begin{aligned} [\hat{J}_z]_{\text{lin}} &= \frac{1}{\sqrt{2}} \begin{pmatrix} \hbar & -\hbar \\ i\hbar & i\hbar \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} \hbar(1) + (-\hbar)(1) & \hbar(-i) + (-\hbar)(i) \\ i\hbar(1) + i\hbar(1) & i\hbar(-i) + i\hbar(i) \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} 0 & -2i\hbar \\ 2i\hbar & 0 \end{pmatrix} \end{aligned}$$

Simplifying entries:

$$[\hat{J}_z]_{\text{lin}} = \begin{pmatrix} 0 & -i\hbar \\ i\hbar & 0 \end{pmatrix} = \hbar \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \hbar \hat{\sigma}_y \quad (5)$$

Physical Interpretation: Invariance of Quantum Observables

The Essential Pedagogical Conclusion

What to Emphasize to Students:

1. **Off-Diagonal Structure:** In the linear basis, $[\hat{J}_z]_{\text{lin}}$ is purely off-diagonal. This mathematical structure immediately confirms that the linear states $|x\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|y\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are *not* eigenstates of angular momentum:

$$[\hat{J}_z]_{\text{lin}}|x\rangle = \hbar \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = i\hbar \begin{pmatrix} 0 \\ 1 \end{pmatrix} = i\hbar|y\rangle \neq \text{const} \cdot |x\rangle$$

2. **Eigenvalue Invariance:** If we diagonalize $[\hat{J}_z]_{\text{lin}}$, its characteristic equation is:

$$\det \begin{pmatrix} -\lambda & -i\hbar \\ i\hbar & -\lambda \end{pmatrix} = \lambda^2 - \hbar^2 = 0 \implies \lambda = \pm\hbar$$

The eigenvalues are identical to those in the circular basis!

3. **Core Conceptual Takeaway:**

**The matrix representation of an operator depends on the basis,
while the underlying physical observable and its eigenvalues are completely invariant.**

Extensions: From Polarization to Quantum Information

Looking Ahead: Quantum Qubits and Bell Nonlocality

- **The Photon as a Quantum Qubit:** Because the polarization space is two-dimensional, photon polarization provides the physical implementation for quantum bits (qubits) in optical quantum computing and quantum key distribution (BB84 protocol):

$$|0\rangle \equiv |x\rangle, \quad |1\rangle \equiv |y\rangle$$

- **Polarization Entanglement:** Spontaneous parametric down-conversion (SPDC) produces photon pairs entangled in polarization:

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|x\rangle_1|y\rangle_2 - |y\rangle_1|x\rangle_2) = \frac{i}{\sqrt{2}}(|R\rangle_1|L\rangle_2 - |L\rangle_1|R\rangle_2)$$

This rotationally invariant singlet state is used directly in experimental tests of Bell's inequalities, proving the non-local nature of quantum reality.

- **Angular Momentum Conservation in Atoms:** When an excited hydrogen atom undergoes a $2p \rightarrow 1s$ dipole transition with magnetic quantum number change $\Delta m = \pm 1$, conservation of angular momentum requires the emitted photon to carry away angular momentum $\pm\hbar$, producing circularly polarized photons.

Formative Assessment & Conceptual Check

Name: _____

Date: _____

Task 1 (Eigenvector Verification): Multiply the matrix $[\hat{J}_z]_{\text{lin}} = \hbar \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ by the vector representation of $|R\rangle$ in the linear basis, $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$, and verify that it is an eigenvector with eigenvalue $+\hbar$.

Task 2 (Measurement Probability): A single photon is prepared in the right-circular polarization state $|R\rangle$. It passes through an ideal linear polarizer oriented at angle θ relative to the x -axis ($|\theta\rangle = \cos\theta|x\rangle + \sin\theta|y\rangle$).

- (a) Calculate the probability amplitude $\langle \theta | R \rangle$.
- (b) Apply the Born Rule to compute the probability of transmission, $P(\theta)$. Does the transmission probability depend on the angle θ ? Explain the physical meaning of your result.

Curricular Context & References

This instructional sample corresponds to Chapter 2 of Townsend's textbook for upper-division quantum mechanics.

- Townsend, J. S. *A Modern Approach to Quantum Mechanics*, 2nd ed.; University Science Books: Mill Valley, CA, 2012; Chapter 2 (Rotation of States and Angular Momentum), Section 2.5–2.6, pp. 61–67.
- Griffiths, D. J.; Schroeter, D. F. *Introduction to Quantum Mechanics*, 3rd ed.; Cambridge University Press, 2018; Chapter 4 (Quantum Mechanics in Three Dimensions).
- Hecht, E. *Optics*, 5th ed.; Pearson, 2017; Chapter 8 (Polarization).